

Are Cryptocurrency Markets Running Behind the Fed?

A significant shift in crypto markets microstructure

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Table of Contents

1. Key Outcomes	3
2. Introduction	4
3. Previous Research	6
4. Diversify but verify	8
4.1. Annualised Rolling Sharpe Ratio for 2021	8
4.2. Asset Classes Reaction Synced after April 2021 Inflation Data Release	9
5. To taper or not to taper?	11
5.1. US job market lagging behind expectations	11
5.2. Doves versus Hawks	12
5.3. Careful Announcement	14
6. Mind the Spread	16
6.1. Seeking Support	16
6.2. The Liquidity Hole	17
7. Conclusion	19
8. References	21
9. Contacts	22

Key Outcomes

We have come to 2 key conclusions that might indicate significant changes in the cryptocurrency market microstructure:

1. In contrast to recent research, we note that **BTC/USD was sensitive to major Fed policy announcements** in Q2-Q3 2021 similar to main asset classes.
2. OTC Liquidity Providers tend to **provide twice as narrow spreads** in comparison to Centralised Crypto Exchanges during market volatility related to macroeconomic news.

Introduction

As institutional investors increasingly think about getting exposure to cryptocurrencies, we hear many different claims regarding the benefits of allocation to such investment vehicles for portfolio managers.

According to some of the recent publications, cryptocurrencies offer attractive Sharpe ratios compared to other asset classes. They are presenting an opportunity for risk diversification, being driven mainly by a news flow related to regulation, crypto-payment adoption or hacking. They can even be used as “safe havens” during periods of market turmoil under economic uncertainty.

In this article, we show that those statements need to be taken with a grain of salt. We could witness the markets riding through a period of indecision from the US Fed in Q2-Q3 2021 when policymakers had to wait on whether to scale down their ultra loose monetary strategy. We find evidence that cryptocurrencies have been no exception to a volatile and collective response across all markets to Fed officials’ commentary and important US macroeconomic data releases missing forecasts during this period.

In the vicinity of Fed announcements, we report increasing trading costs across the main crypto-exchanges including Coinbase, Binance, Kraken and Bitstamp. We also compare the quality of

liquidity on centralised venues with firm aggregated liquidity of global OTC desks: BlockFills, Cryptology, DV Chain, oneAlpha and Veliona on wholesale marketplace Finery Markets. That helped us to investigate the predictability of execution quality under such scenarios.

Previous Research

It has been well documented before that loose monetary policy and any form of quantitative easing from central bankers fuel enthusiasm for investment into decentralised currencies such as Bitcoin. This brings volatility to those markets, especially when decisions are awaited from policymakers. Corbet et al. (2017), studied the impact of QE announcements from major central banks, including US, UK, EU and Japan. They considered daily returns for the BTC/USD rate, from its inception up to 2016, and found that policy action had a non-trivial impact on its volatility. This work has been our main inspiration.

Š. Lyócsa et al. (2020), rely on intraday returns for BTC/USD from Bitstamp between Jan 2013 and Dec 2018. They build a volatility model which is completely overwhelmed by shocks due to news on cryptocurrency regulation, press sentiment and hacking attacks on crypto-exchanges. They hardly manage to capture any sensitivity to the macroeconomics calendar, apart from forward-looking surveys. They end up ultimately claiming that no influence from major index readings and policy announcements could be determined.

The work of M. Schnaubelt et al. (2019), gathers stylistic facts about the limit order book for BTC/USD from Dec 2017 to Nov 2018. They look at several exchanges and provide a comparison with other markets.

We contribute to the existing literature in several ways. First of all, we analyse both intraday and market microstructure data to study the behaviour of the BTC/USD rate around macroeconomic releases and policy announcements. We investigate several exchanges and compare them to OTC liquidity providers on Finery Markets. Finally, we focus on the latest developments in Q2-Q3 2021, with the US economy living through a bumpy recovery from the coronavirus pandemic, and the US Fed dealing with a challenging set up involving a struggling job market on one hand and record-beating inflation on the other.

Diversify but verify

In May, BTC/USD erased over 39% of its USD value just under two weeks

Annualised Rolling Sharpe Ratio for 2021*



Fig. 1: Even though the Sharpe ratio penalises upward volatility, this single risk metric overstated Bitcoin as an attractive investment opportunity for the first half of 2021. In May, it erased over 39% of its USD value in just under two weeks.

The lessons learned from Corbet et al. (2017) seem to have been forgotten in recent literature. Excited by talks from large institutions about the possibility to get some exposure to crypto-markets, many studies such as E. Platanakis et al. (2020) and D. T. Pele et al. (2021) rush to overestimate the risk adjusted returns and underestimate the correlations when comparing cryptocurrencies to other asset classes.

Such an approach poses serious problems to risk management. Any investor that considers exposure to cryptocurrencies for its portfolio should reevaluate the claims made above.

* based on data collected from tradingeconomics.com

In this paper, we will illustrate the concerns we have with a series of charts and examples. We will demonstrate that the portfolio manager needs to run its tactical allocation strategy under stress scenarios when no diversification assumption is possible for its cryptocurrency exposure. In fact, the losses across asset classes could be even amplified for the latter.

An inflation release triggered a correlated response from the main markets including BTC/USD

Asset Classes Reaction Synced after April 2021 Inflation Data Release*

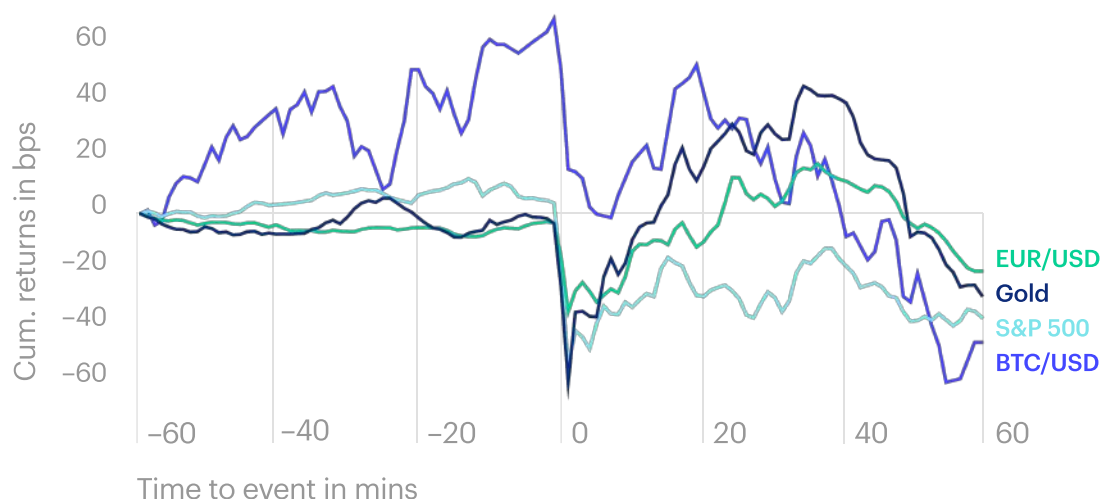


Fig. 2: On May 12th at 12:30 UTC, the U.S. Bureau of Labor Statistics released the figure 4.2% for the annual inflation rate in April, well above markets' forecasts of 3.6%, triggering a correlated response from the main markets, including BTC/USD.

Coming to the second quarter of 2021, with the Fed involved in an unprecedented asset purchasing scheme to support the US economy during the coronavirus pandemic, risks from inflationary

* based on data collected from tradingeconomics.com

pressure started to build up. This was mainly due to disruptions in supply chains and soaring commodity prices.

But the market raised the question whether policymakers will remain ultra-accommodative for long. Those concerns translated into a US stock market sell-off in the run-up to the inflation reading for April.

This coincided with a large turmoil in the cryptocurrency markets. Of course, we cannot assert that the two are necessarily connected. But the coordinated intraday response to the negative surprise reading for the inflation rate that one can observe for various asset classes and the BTC/USD rate, leaves us in no doubt. The prospects of a Fed tapering its ultra-loose monetary policy sooner than expected played a non-trivial role in the cryptocurrencies' May rollercoaster.

To taper or not to taper?

The NFP release increased the odds against tapering and triggered volatility for BTC/USD

US job market lagging behind expectations*

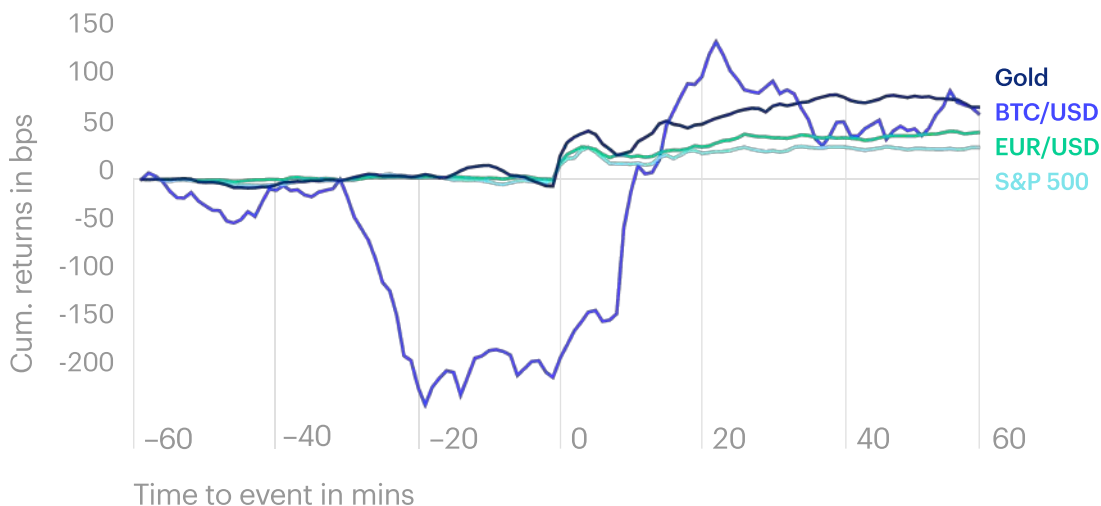


Fig. 3: On June 4th at 12:30 UTC, the U.S. Bureau of Labor Statistics announced that 559K jobs were added in May, well below the forecasted figure of 650K, increasing the odds against tapering and triggering volatility for BTC/USD.

Whenever there is considerable uncertainty in the central bank's policy, market participants pay very close attention to the main macroeconomic indicators used by officials in their decision-making process. Inflation and employment related indices are obviously some of the key figures in this case.

In the US, Non-Farm Payrolls data is an important monthly indicator on which the Fed is typically relying, amongst other signals, to judge the quality of the US economy, and in particular, its recovery from the coronavirus pandemic, as far as our study is concerned.

* based on data collected from tradingeconomics.com

Typically, the markets would expect a certain figure release and would price in its consequences on the policy. There might be, however, high uncertainty involved with the forecast and this could cause volatility already in the run up to the reading. We have seen the above with the turmoil caused by the US inflation news in May.

The FOMC minutes triggered a cold reaction across the main markets, including BTC/USD

Doves versus Hawks*

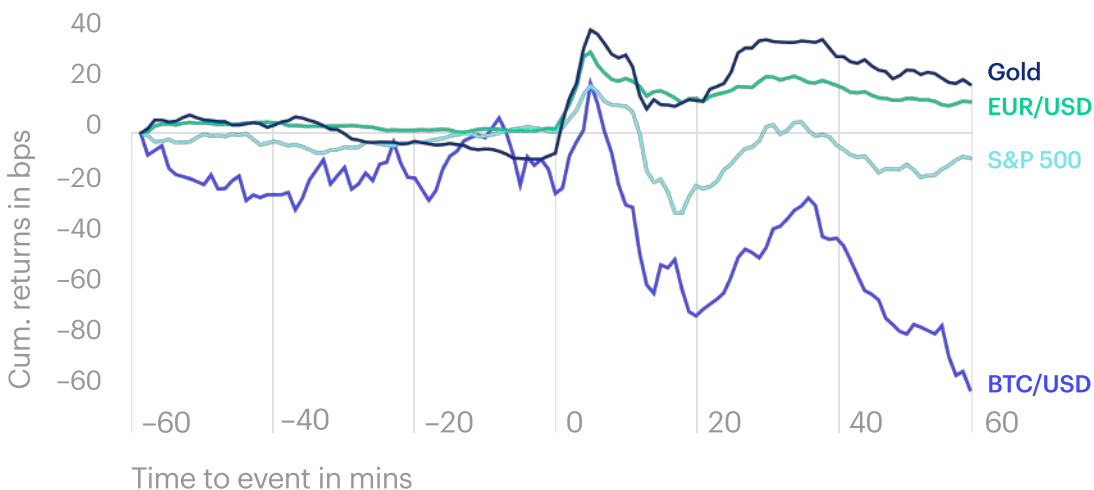


Fig. 4: On August 18th, 18:00 UTC, in the FOMC minutes, officials supported reducing stimulus measures by the end of 2021. But they disagreed on the scale of tapering given the unsatisfactory progress towards full-employment in the US, triggering a cold reaction across the main markets including BTC/USD.

The release itself might also miss the forecast, in a direction which could impact the policy. This happened with the inflation reading in May, but also with employment data in June. The latter raised questions whether the economy is strong enough for the Fed to start tapering in the near future.

* based on data collected from tradingeconomics.com

As we can see with the NFP release, such new information increased the upside risk across several instruments, cryptos included.

The Fed tries to avoid any surprises in its announcements and seeks maximal transparency. Its representatives give regular speeches helping the market to understand their take on the current situation in the economy and provide forward looking criteria for their action on policy.

Summer 2021, however, provided a challenge for the Fed in that respect. The record-beating inflation readings, showing little signs of being just an ephemeral effect, gave a strong case to the critics of the large asset-purchasing scheme in place.

The Jackson Hole announcement was cheered by the main markets, BTC/USD in particular.

Careful Announcement*

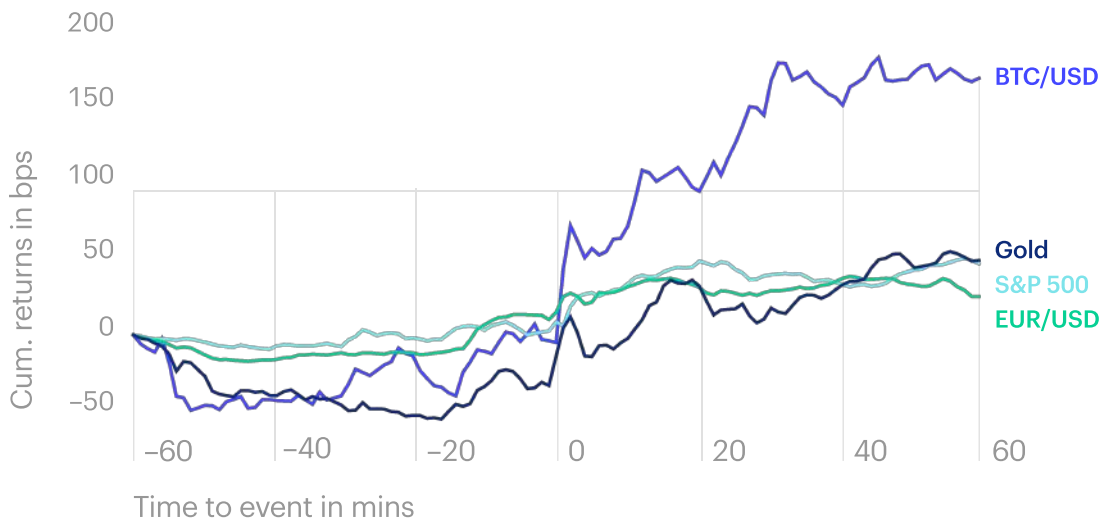


Fig. 5: On August 27th, 14:00 UTC, at the Jackson Hole symposium, Fed Chair, Jerome Powell, essentially confirmed that tapering will start by the end of the year, but struck a dovish tone on raising interest rates anytime soon. That was cheered up by the main markets, BTC/USD in particular.

But the weak progress in US employment, low business confidence and, more importantly, the risk presented by the renewed spread of coronavirus infections couldn't simply untie policymakers' hands.

Given such a window of hope for further stimulus, the support for tapering at the FOMC meeting in August left the markets in the cold. However, the confirmation that followed from the Fed Chair at the Jackson Hole summit came with reassurances. The scale of the tapering and any potential rate hikes were strictly conditioned on strong progress in the economy. That was cheered by all market participants.

* based on data collected from tradingeconomics.com

In both of these announcements, we can distinguish very similar patterns in the response from all representatives of the different asset classes we picked up for illustration. This means, from an academic point of view, that any methodology that classifies any of the classes as impacted by the events above, will have to classify the rest as such too. In particular, this holds for the BTC/USD rate which is what we wanted to demonstrate.

Mind the Spread

BTC/USD tried to find support but broke through at FOMC minutes

Seeking Support*

10 BTC/USD Bid-Ask Spread & Commission** to Mid-Price ratio in bps

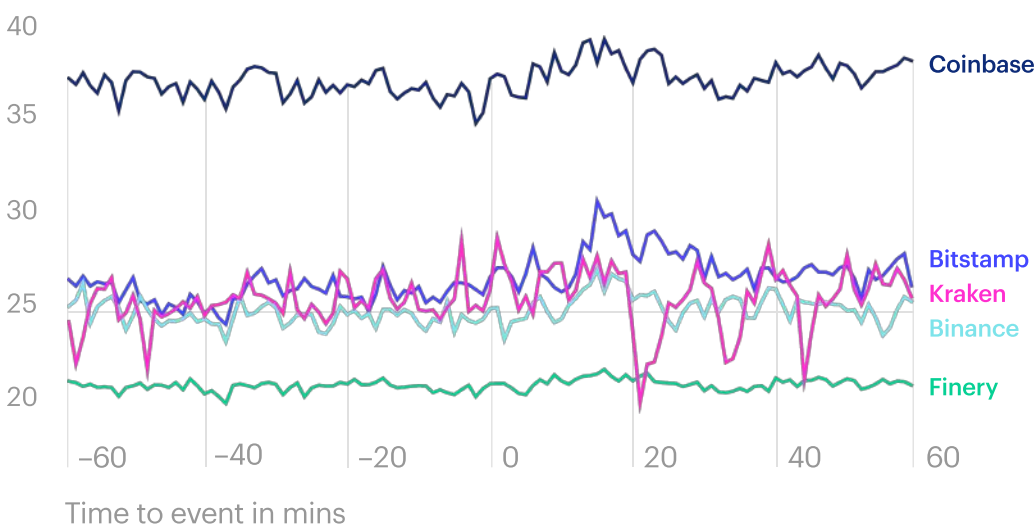


Fig. 6: On August 18th, 18:00 UTC, at the FOMC minutes (cf. Fig. 4), BTC/USD tried to find support but broke through. This caused some widening and higher variation for bid-ask spreads on all exchanges. On Finery Markets however, those effects were minimal.

We provide further evidence that BTC/USD was sensible to recent Fed policy decisions by looking at the market microstructure. More precisely we consider the dynamics of the bid-ask spread during Fed commentary from August 18th & 27th. We compute the spread based on 10 Bitcoins and look at its ratio to the mid-price, taking a time-weighted average for every minute interval.

Regime-changing announcements from central banks lead to a collective response from the main markets. Most participants see

* data collection platform: github.com/grinisrit/noa-atra

** based on a taker volume of 100M USD p. m

the need to rebalance their portfolio or regauge their strategy.

During the meeting, headlines arrived from the Fed, unrolling its propositions for policy action as well as updates on forward-looking decision criteria. This creates a demand for liquidity under uncertain market conditions. The reason for choosing an amount as large as ten coins is to investigate to what extent market-makers are willing to provide a counterparty for a potentially large trade under such intensive market moving news flow.

Bull rush for BTC/USD at tapering announcement

The Liquidity Hole*

10 BTC/USD Bid-Ask Spread & Commission** to Mid-Price ratio in bps

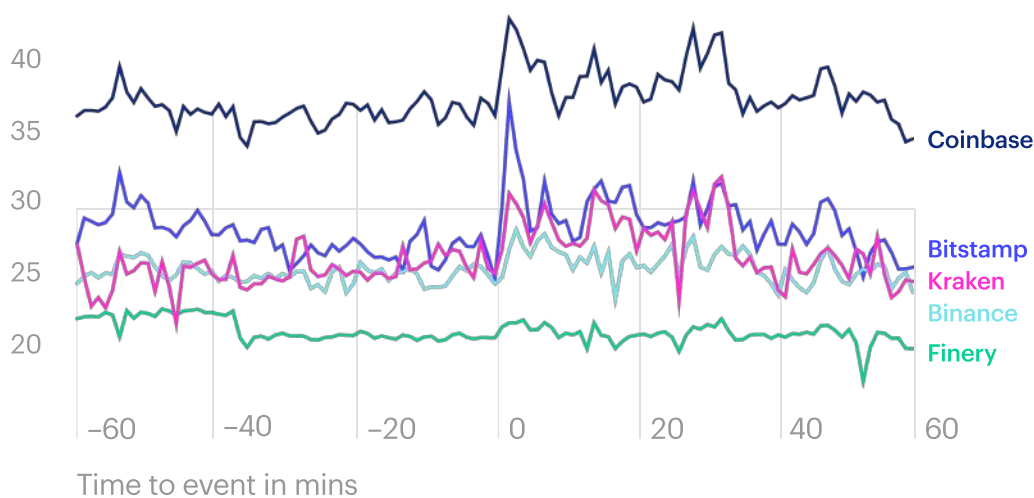


Fig. 7: On August 27th, 14:00 UTC at the Jackson Hole summit (cf. Fig. 5) BTC/USD went through a bull rush with large trading costs deterioration across all exchanges. On Finery Markets, a relatively stable execution quality was provided.

* data collection platform: github.com/grinisrit/noa-atra

** based on a taker volume of 100M USD p. m.

It is well documented for the main markets such as FX, stocks, commodities or bonds that these conditions lead to market makers asking for an extra execution premium. The work of N. Hautsch et al. (2017), for example, presents a very detailed study in this direction for Eurex Bund Futures.

In our case, we expect widening and an increase in volatility for the BTC/USD bid-ask spread. For both events, we find that this hypothesis is largely confirmed. We note that, for the Fed Chair, Jerome Powell's, speech on August 27th, a particularly severe liquidity premium shock was incurred across all exchanges and trading venues in our sample.

As a final note, we believe that the facts we presented in this section should be taken into consideration when designing a risk management system & execution strategy for portfolios involving cryptocurrency exposure. Any investment strategy into that asset class must be examined thoroughly under stress tests involving high correlation with the main market components, spiking volatility and rise in trading costs.

Conclusion

We have studied the BTC/USD rate response to US Fed monetary policy news flow from Q2-Q3 2021. In comparison with other asset classes, we found a coordinated intraday response to surprise readings for the inflation rate, non-farm payrolls data and any uncertainty stemming from Fed officials' comments.

Throughout August 2021, we scrutinised level II quotes from the main crypto-exchanges including Coinbase, Binance, Kraken and Bitstamp. We showed evidence for degradation in trading costs, in particular widening bid-ask spreads, during Fed meetings where major monetary policy announcements were made. We also examined the firm quotes posted by OTC liquidity providers BlockFills, Cryptology, DV Chain, oneAlpha and Veliona on Finery Markets. In comparison to the other crypto-trading venues we considered, we report a more predictable execution quality there throughout the investigated events.

We believe that our findings will help a portfolio manager seeking exposure to cryptocurrency markets to develop a robust risk management system alongside with proper trading setup for its tactical allocation strategy.

Looking forward, this is just the beginning of a regime change for the US Fed policy. Tapering is yet to start, its scale is conditioned on

the US economy recovery and soon we will be learning about the path policy makers will follow for the actual rate hikes. Therefore we have all the reasons to get excited about further fascinating research on interactions between cryptocurrency markets and monetary policy news flow.

Acknowledgements:

We kindly acknowledge support from [Finery Markets](#), [GrinisRIT](#) and the [JetBrains Research Foundation](#)

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